

ECEN 607: Advanced Analog Circuit Tech

Lab 3: Op Amp Design - I

Pre-lab or Lab Report

Name: FirstName LastName

UIN: ***Last Four Digits**

Objectives

- To be able to identify the dominant noise sources.
- To study Miller compensation with RHP zero cancellation techniques.
- To study and design a single-ended telescopic op amp.

Recommendation:

Start building common testbench setups for finding op amp parameters such as DC Gain, unity gain frequency, slew rate etc. This can be re-used for different op-amps that you will be designing in this course and later, hence saving a lot of time.

Prelab

- **In this prelab, 1.8V NFETs and PFETs will be used.**
- **Do not forget to show a detailed design procedure for each design.**
- **The following op amps must be tested in unity feedback gain configuration. YOU MAY WANT TO REPLACE M9 AND M10 BY A SIMPLE RESISTOR.**

1. There are various techniques to remove or move the RHP zero that comes with the Miller compensated classical two-stage opamp. One of the techniques is to place a resistor in series with C_c . This resistor can be implemented with two transistors operating in triode region, as shown in Fig. 3-1 OR A SINGLE PASSIVE RESISTOR. Find the proper transistor dimensions and check the bias conditions.

Hint: be sure that $100\text{mV} < V_{\text{DSAT}} < 250\text{mV}$.

a) Design the circuit shown in Fig 3-1 to obtain the specifications given (Table 3-1). Identify the dominant noise sources (**leave the topic a) pending, go to item b)**) and find the expression for the input referred thermal noise density. Show the hand-calculations for the input referred thermal noise density and report the estimated value. Obtain the profile of the flicker noise from simulations and identify the corner frequency between thermal and flicker noise density.

b) Estimate the compensating resistor to cancel the RHP zero and be sure you guarantee phase margin of 45° or better. One can become greedy and decide to move the zero and cancel the first non-dominant pole. Compare the magnitude and phase response of the circuit and report phase and gain margin values. Show screen shots. Be sure the plots have enough sensitivity, and font sizes are visible.

c) Over design the RHZ by moving the zero to the left hand side of the S-plane around the value of gm_8/C_c . Repeat the simulations in b) and compare the results with the previous case.

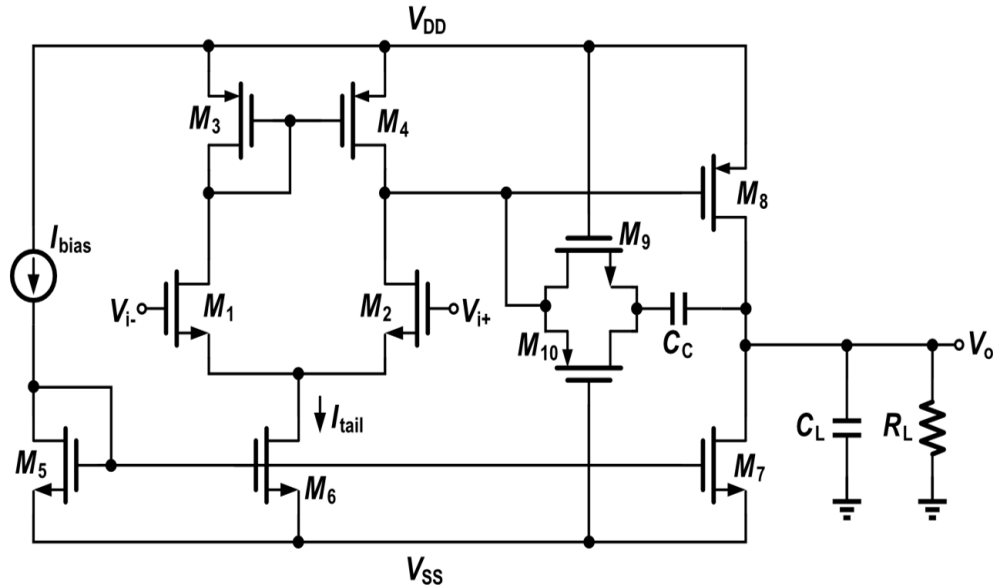


Figure 3-1: Two-stage amplifier with Miller compensation and RHP zero cancellation

Table 3-1: Design specifications

$V_{DD}-V_{SS}$	1.8V
A_{v0}	> 40 dB
GBW	> 2 MHz
PM	> 45°
Output Swing	> 1 V
C_L	20 pF
R_L	10k Ω
Input referred noise (10 Hz – 2 MHz)	< 30 μ Vrms (ignore the noise in this lab)

Lab Report

Design:

1. Simulate the circuit from the prelab and adjust the transistor sizes accordingly until all specifications are met. Provide necessary screen shots such that you clearly show the specs are met.
2. Run brute-force MC simulations and update the best & worst corners that you created in Lab1 with information from best and worst 3-sigma noise values.
3. Report the DC Gain, GBW, Phase Margin and Gain Margin for all corners. (No plots needed, just report the numbers in a comprehensive table).
4. Adjust your design to target a 3-sigma yield for DC Gain, GBW, Phase Margin (Show histograms).
5. Do a cross simulation (two corners, supply variation and temperature variation) and report the worst case DC Gain, GBW, Phase Margin and input referred noise. Assume there is +/- 5% variation on the supply, -40 < temperature < 125 C. You do not have to adjust the design if the specs are not met. Provide comparison tables of hand-calculated vs. final transistor sizes, and required specs vs. simulated specs.
6. Comment on your learning from this lab.